Onde orografiche: onde di gravità se aria stabilmente stratificata e vento impatta su ostacolo. Dipende da param geometrici hill, h e larg metà altez b. Vento, se intenso e
se stabile stratificato, dth/dt > 0. h' = N h / U se h' << 1 => appross onde lineari. b' = N b / U come onde si propagano => b' >> 1 regime idrostatico, quindi prop onda principalmente verticale. Ora simuliamo regime lineare idrostatico, c'è soluzione analitica, con solo lungh d'onda verticale lambda = 2 pi U / N, energia trasportata solo verticalmente.
Altrimenti onde di Lee.

Idealized numerical simulations of a 2D flow over a bell-shaped hill

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Laboratory of Complex Terrain Meteorology course

Prof. L.S. Leo

Bologna, 11/12/2025

Objectives

The aim of this laboratory experience is to study the behavior of flows over complex terrain.

We focus on the case of a 2D flow with constant wind over a small ridge, in different configurations of the atmospheric stability, and varying wind velocity and hill height.

We will do that performing idealized numerical simulations using the Weather Research and Forecasting (WRF) model.

Objectives

By the end of this laboratory, you will be able to:

- Familiarize with the main physical mechanisms controlling flow over orography;
- Gain practical experience in configuring, compiling, and running idealized WRF cases;
- Visualize and interpret model outputs.

Weather Research and Forecasting (WRF) model

A state-of-the-art mesoscale numerical weather prediction system designed for both atmospheric research and operational forecasting applications.
The model serves a wide range of meteorological applications across scales from tens of meters to thousands of kilometers.
For researchers, WRF can produce simulations based on actual atmospheric conditions (i.e., from observations and analyses) or idealized conditions.

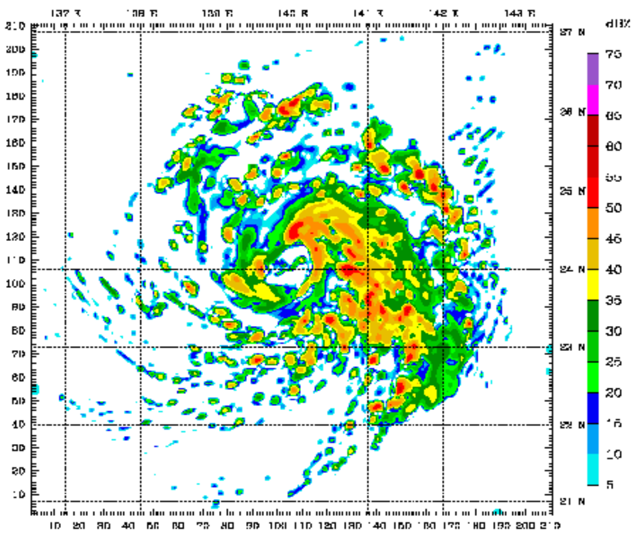
Current version is 4.6

Introduction

Weather Research and Forecasting (WRF) model

What can WRF be used for?

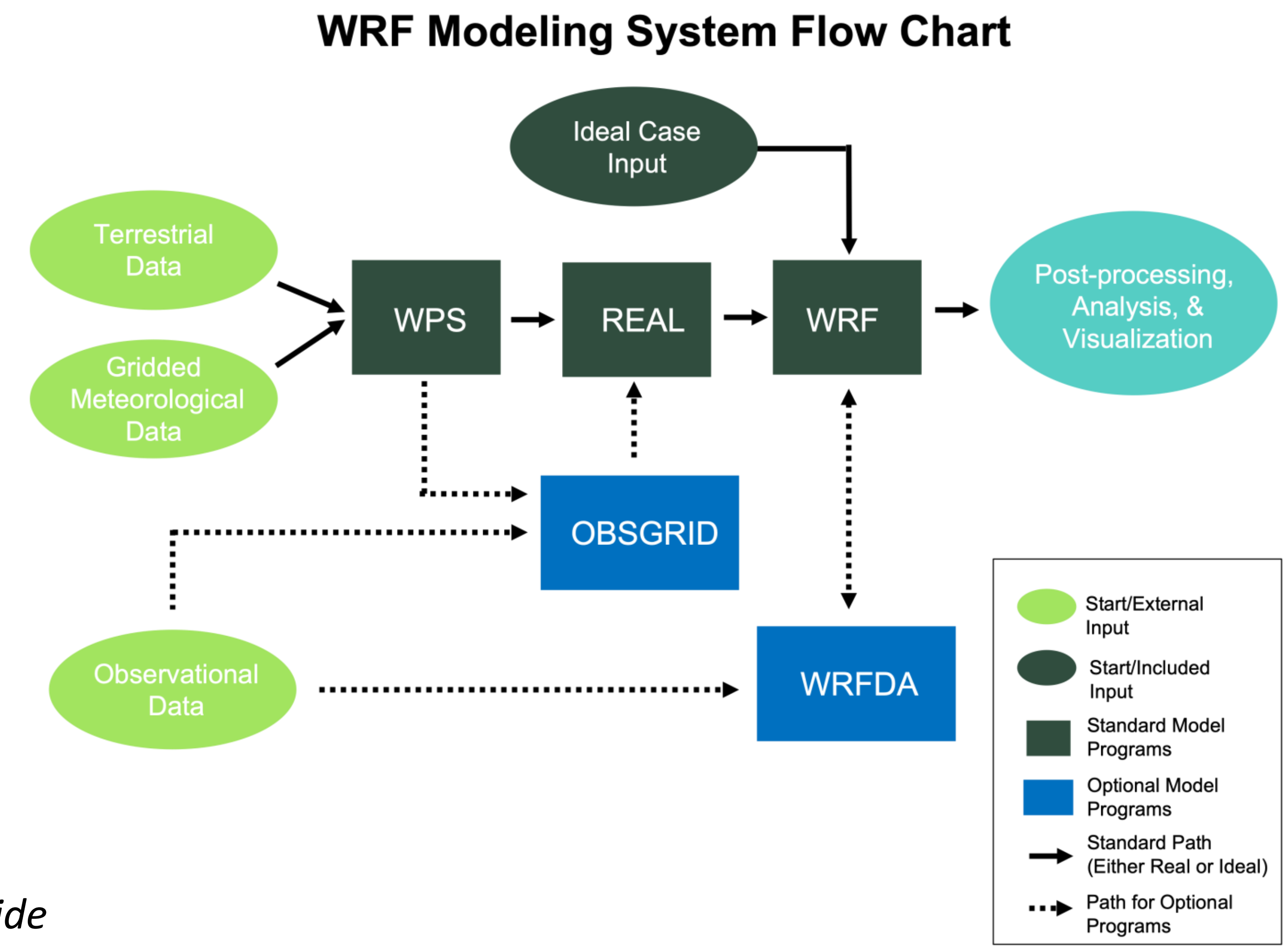
- Atmospheric physics/parameterization research
- Case study research
- Real-time NWP and forecast system research
- Regional climate and seasonal time-scale research
- Coupled chemistry applications
- Global simulations
- Idealized simulations at various scales (convection, baroclinic waves, LES)



@ Atmoz

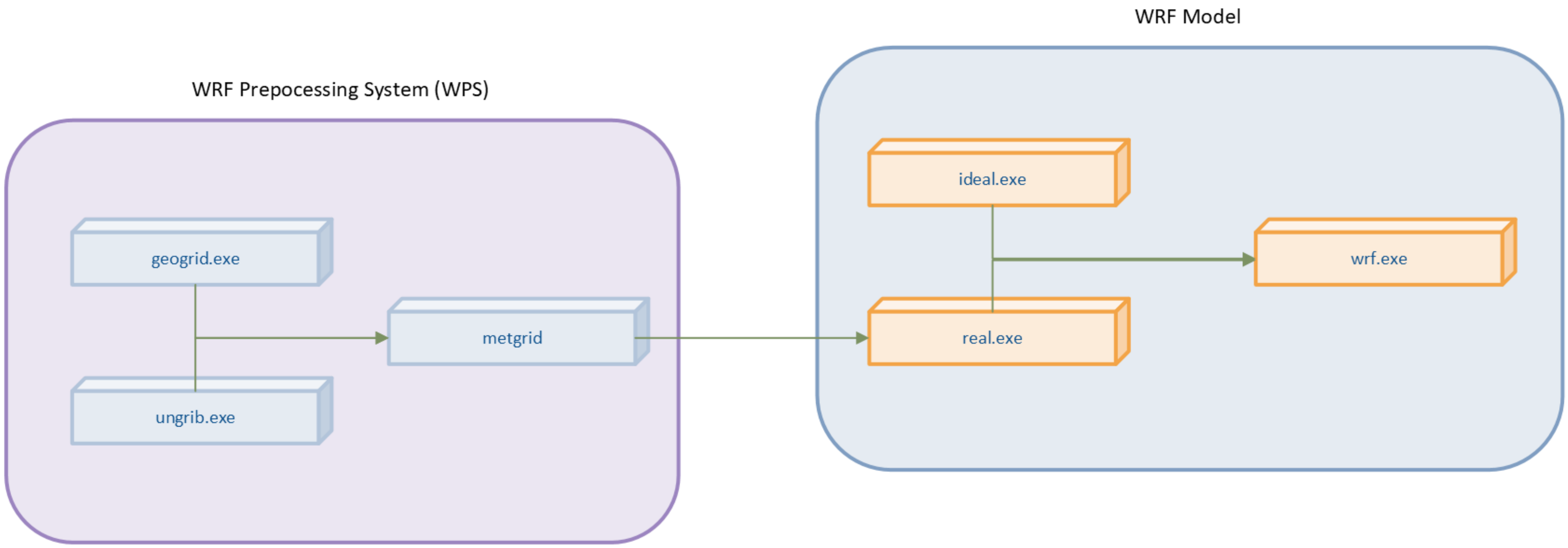
Introduction

• Flow over a hill • WRF Model



From WRF User Guide

WPS and WRF Program Flow



Introduction

- The equation set for ARW is **fully compressible, Eulerian and nonhydrostatic** with a run-time hydrostatic option.
- It is conservative for scalar variables.
- The model uses terrain-following, **hybrid sigma-pressure vertical coordinate** with the top of the model being a constant pressure surface.
- The horizontal grid is the **Arakawa-C grid**.
- The time integration scheme in the model uses the third-order Runge-Kutta scheme, and the spatial discretization employs 2nd to 6th order schemes.
- The model supports both idealized and real-data applications with various lateral boundary condition options.
- The model also supports one-way, two-way and moving nest options.

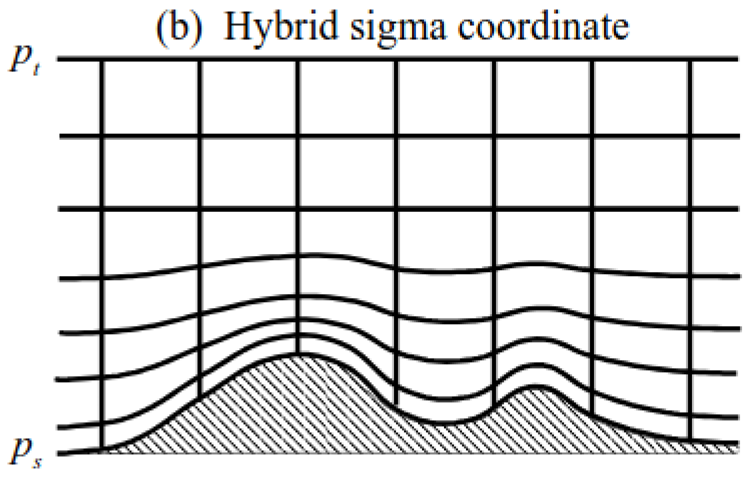
Introduction

• Flow over a hill • WRF Model: Vertical coordinates

The ARW v4 uses **hybrid sigma-pressure vertical coordinate** defined as

$$p_d = B(\eta)(p_s - p_t) + [\eta - B(\eta)](p_0 - p_t) + p_t,$$

p_0: reference sea-level pressure.
$B(\eta)$: defines the relative weighting between the terrain-following sigma coordinate and a pure pressure coordinate, such that η corresponds to the sigma coordinate for $B(\eta) = \eta$ and reverts to a hydrostatic pressure coordinate for $B(\eta) = 0$. To smoothly transition from a sigma coordinate near the surface to a pressure coordinate at upper levels, $B(\eta)$ is defined by a third order polynomial.



ARW coordinate

The ARW v4 uses **hybrid sigma-pressure vertical coordinate.**

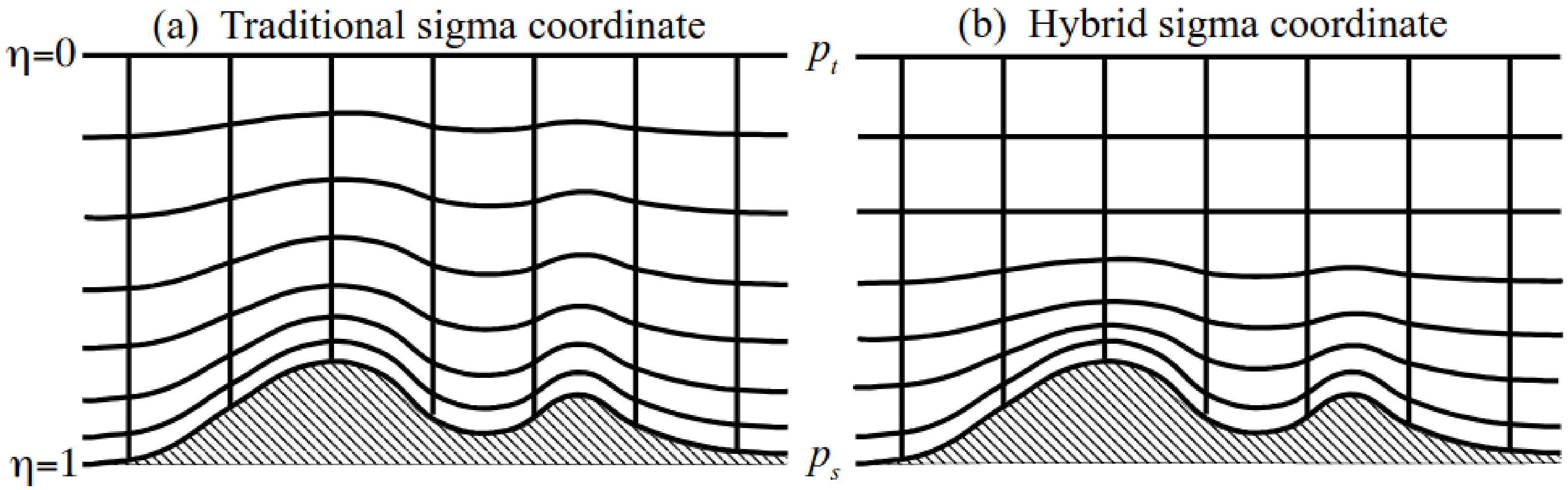


Fig. 2.1 of WRFv4 Technical notes (see references)

Introduction

• Flow over a hill • WRF Model: Equations

WRF ARW solves the **fully compressible, non-hydrostatic equations of motion** with a terrain-following hydrostatic pressure coordinate system, written in flux form. For simplicity, using Cartesian coordinates and neglecting the Coriolis effect, the WRF model solve the followings:

Equation of state

$$p = \rho R_d T;$$

Mass conservation

$$\frac{\partial \rho}{\partial t} + \frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} + \frac{\partial W}{\partial z} = 0;$$

Conservation of momentum

$$\frac{\partial U}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial x} = -\frac{\partial U u}{\partial x} - \frac{\partial V u}{\partial y} - \frac{\partial W u}{\partial z} + F_x,$$

$$\frac{\partial V}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial y} = -\frac{\partial U v}{\partial x} - \frac{\partial V v}{\partial y} - \frac{\partial W v}{\partial z} + F_y,$$

$$\frac{\partial W}{\partial t} + c_p \Theta \frac{\partial \pi}{\partial z} + g\rho = -\frac{\partial U w}{\partial x} - \frac{\partial V w}{\partial y} - \frac{\partial W w}{\partial z} + F_z;$$

Conservation of energy

$$\frac{\partial \Theta}{\partial t} + \frac{\partial U\theta}{\partial x} + \frac{\partial V\theta}{\partial y} + \frac{\partial W\theta}{\partial z} = \rho Q.$$

$$U = \rho u, \quad V = \rho v, \quad W = \rho w, \quad \Theta = \rho\theta,$$

(u, v, w) are the velocity components
θ is the potential temperature
T is the absolute temperature
π is the Exner function
F_x, F_y, F_z are the friction terms

A more detail description in WRFv4 Technical notes

The spatial discretization in the ARW solver uses a **C grid staggering** for the variables.

Normal velocities are staggered one-half grid length from the thermo-dynamic variables.
The variable indices, (i, j, k) indicate variable locations with $(x, y, \eta) = (i\Delta x, j\Delta y, k\Delta\eta)$.

u, v, and w are defined as u points, v points, and w points, respectively, while points where θ is located as mass point.

Moisture variables q_i and the coordinate metric μ are defined at the mass points on the discrete grid, while the geopotential ϕ that is defined at the w points.

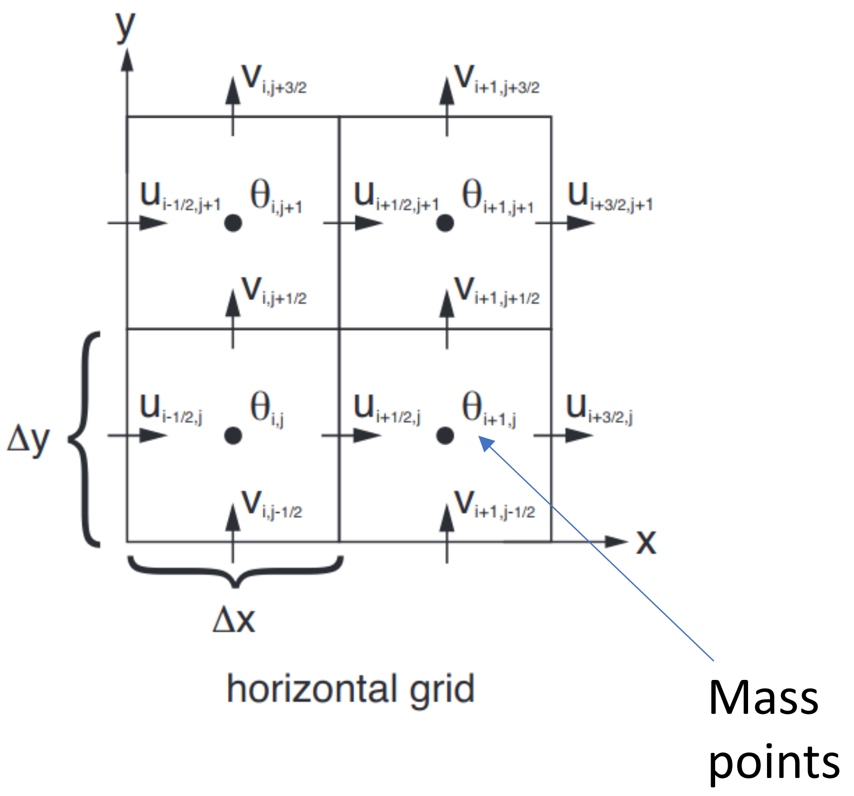


Fig. 3.2 of WRFv4 Technical notes

Introduction

The spatial discretization in the ARW solver uses a C grid staggering for the variables.

The vertical grid length Δη is not a fixed constant, it is specified in the initialization. The user is free to specify the η values of the model levels subject to the constraint that η = 1 at the surface, η = 0 at the model top, and η decreases monotonically between the surface and model top.

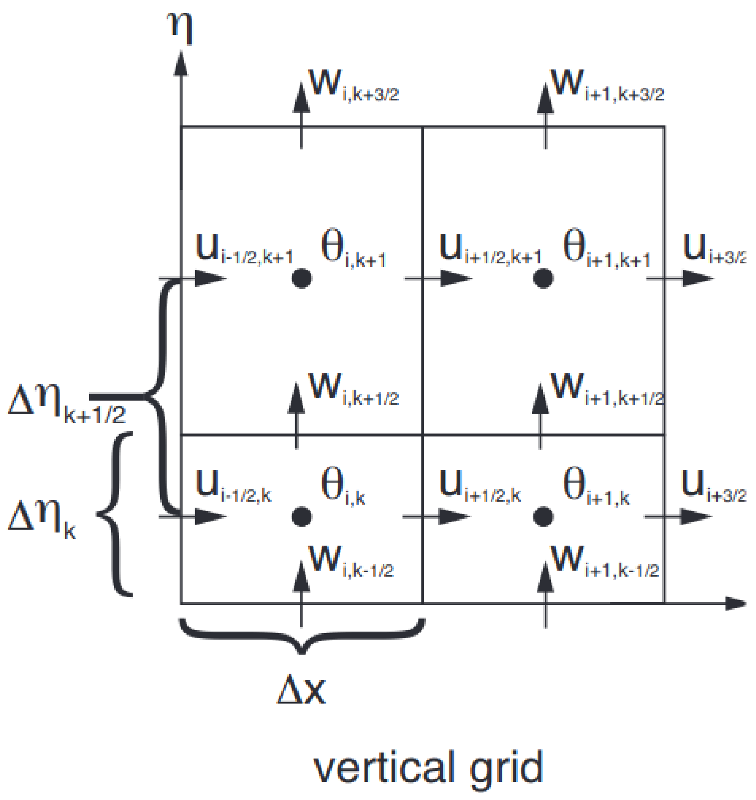
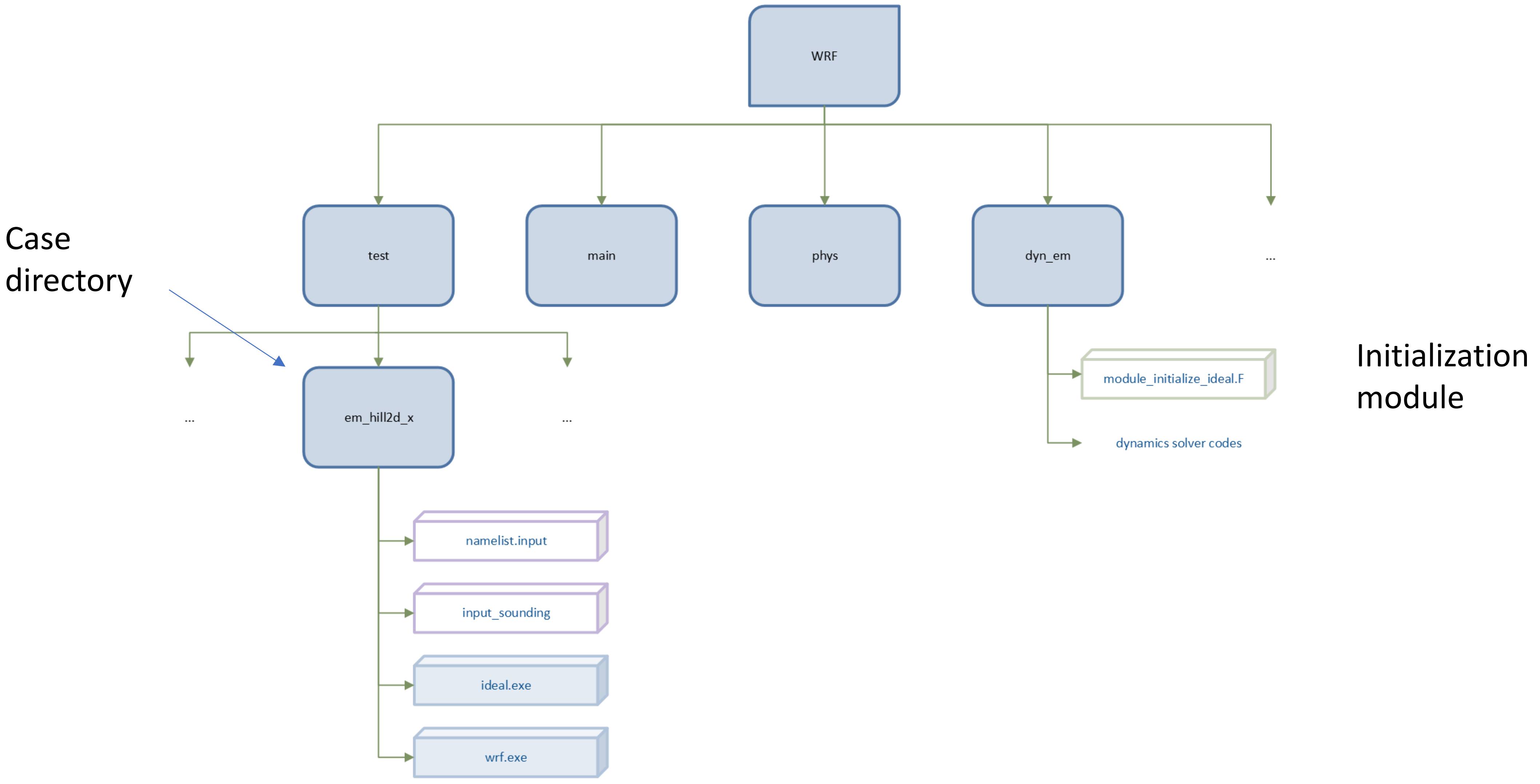


Fig. 3.2 of WRFv4 Technical notes

Introduction

• Flow over a hill • WRF Model: 2D hill simulation



Introduction

Let's now see the case we will work on.

Steps:

1) Access the cluster
2) Set the environment: create a conda environment and install Ncview to view wrf outputs
3) Configure and compile WRF
4) Run the case

Introduction

• Flow over a hill • WRF Model

Access the cluster:

```
ssh -X -J nome.cognome@137.204.165.34 nome.cognome@137.204.165.41
```

Download WRF:

```
git clone --recurse-submodules https://github.com/wrf-model/WRF
```

To compile WRF use an interactive job:

```
salloc -N 1 --cpus-per-task=4 --time=00:30:00 --mem=32G --
  constraint=blade --partition=b1
srun --pty --overlap --jobid $JOBID bash
```

Then, set the environment:

Introduction

• Flow over a hill • WRF Model: Configuration and Compiling

```
tcsh
setenv DIR /home/software/atmos/WRFv4/Build_WRF/LIBRARIES
setenv CC gcc
setenv CXX g++
setenv FC gfortran
setenv FCFLAGS -m64
setenv F77 gfortran
setenv FFLAGS -m64
setenv JASPERLIB $DIR/grib2/lib
setenv JASPERINC $DIR/grib2/include
setenv LDFLAGS -L$DIR/grib2/lib
setenv CPPFLAGS -I$DIR/grib2/include
setenv PATH $DIR/netcdf/bin:$PATH
setenv NETCDF $DIR/netcdf
setenv LD_LIBRARY_PATH /home/software/atmos/WRFv4/Build_WRF/LIBRARIES/grib2/lib
setenv LD_LIBRARY_PATH $DIR/grib2/include:$LD_LIBRARY_PATH
module load mpi/openmpi/4.1.4
```

Introduction

• Flow over a hill • WRF Model: Configuration and Compiling

In the WRF main directory type:

```
./configure
```

For idealized simulations, choose option 32 with no nesting (0)

Open `configure.wrf` file and modify line 137:

```
vi configure.wrf
```

type `:137` and press enter. Press `i` to go in `INSERT MODE` and delete “`time`”.

To exit press `esc` and save (`:wq`).

These commands refers to Vim text editor, but you can choose whatever you prefer (e.g. GNU nano)!

Before compiling, change hill height (see next slide).

Now you can compile:

```
./compile em_hill2d_x >& compile.log
```

Call at the end:

```
scancel $JOBID
```

If compilation is successful, you should see executables in the main directory. In WRF folder, type:

```
ls main/
```

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

dyn_em/module_initialize_ideal.F

```
 CALL model_to_grid_config_rec ( grid%id , model_config_rec , config_flags )

ideal_constants: SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
 hm = 100.
 xa = 5.0

 icm = ide/2


 xa1  = 5000./500.
 xal1 = 4000./500.
 pii  = 2.*asin(1.0)
 hm1  = 250.
  hm1  = 1000.


 stretch_grid = .true.
 delt = 0.
  z_scale = .40
 z_scale = 8000./config_flags%ztop
 pi = 2.*asin(1.0)
 write(6,*) ' pi is ',pi
 nxc = (ide-ids)/2
 nyc = (jde-jds)/2
```

hill height (m) and half-width (grid points)

hill position in domain (center gridpoint in x)

In vim:
Type `/hill`
Press `n` (`N`) to move forward (back) in the search

Introduction

• Flow over a hill • WRF Model: 2D hill simulation

`dyn_em/module_initialize_ideal.F`

```
 CALL model_to_grid_config_rec ( grid%id , model_config_rec , config_flags )

ideal_constants: SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
 hm = 100.
 xa = 5.0

 icm = ide/2


 xa1   = 5000./500.
 xal1 = 4000./500.
 pii   = 2.*asin(1.0)
 hm1   = 250.
  hm1   = 1000.


 stretch_grid = .true.
 delt = 0.
  z_scale = .40
 z_scale = 8000./config_flags%ztop
 pi = 2.*asin(1.0)
 write(6,*) ' pi is ',pi
 nxc = (ide-ids)/2
 nyc = (jde-jds)/2
```

hill shape

lower bc

```
  ideal_terrain: SELECT CASE ( model_config_rec%ideal_case )
  CASE (hill2d_x)
  DO j=jts,jte
  DO i=its,ite
    grid%ht(i,j) = hm/(1.+(float(i-icm)/xa)**2)
!!    grid%ht(i,j) = hm1*exp(-(( float(i-icm)/xa1)**2))   &
!!                   *( (cos(pii*float(i-icm)/xal1))**2 )
    grid%phb(i,1,j) = g*grid%ht(i,j)
    grid%php(i,1,j) = 0.
    grid%ph0(i,1,j) = grid%phb(i,1,j)
  ENDDO
  ENDDO
```

In vim:
Type `/hill`
Press `n` (`N`) to move forward (back) in the search

Witch of Agnesi

$$h(x) = \frac{h_m}{(x/b)^2+1}$$

Gaussian shape

Introduction

• Flow over a hill • WRF Model: 2D hill simulation

`namelist.input`

Section (&)

Simulation time

Frequency (in min) that data are written in the wrfout* file

number of history output times in each wrfout* file

```
&time_control
run_days                            = 0,
run_hours                           = 10,
run_minutes                         = 0,
run_seconds                         = 0,
start_year                          = 0001,
start_month                         = 01,
start_day                           = 01,
start_hour                          = 00,
start_minute                        = 00,
start_second                        = 00,
end_year                            = 0001,
end_month                           = 01,
end_day                             = 01,
end_hour                            = 10,
end_minute                          = 00,
end_second                          = 00,
history_interval                    = 60,
frames_per_outfile                  = 1000,
restart                             = .false.,
restart_interval                    = 1440,
io_form_history                     = 2
io_form_restart                     = 2
io_form_input                       = 2
io_form_boundary                    = 2
/
```

More on namelist variables in
[WRF UG section on Namelist variables]

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

namelist.input

Time step (s)

s: start

e: end

```
&domains
time_step                    = 20,
time_step_fract_num          = 0,
time_step_fract_den          = 1,
max_dom                      = 1,
s_we                         = 1,
e_we                         = 202,
s_sn                         = 1,
e_sn                         = 3,
s_vert                       = 1,
e_vert                       = 41,
dx                           = 2000,
dy                           = 2000,
ztop                         = 30000.,
/
```

Number of grid points

Number of vertical levels

Resolution in metres

Height of the top in metres

```
&bdy_control
periodic_x                   = .false.,
symmetric_xs                 = .false.,
symmetric_xe                 = .false.,
open_xs                      = .true.,
open_xe                      = .true.,
periodic_y                   = .true.,
symmetric_ys                 = .false.,
symmetric_ye                 = .false.,
open_ys                      = .false.,
open_ye                      = .false.,
/
```

Open boundaries (x)

Periodic bc in y

Introduction

- Flow over a hill • WRF Model: 2D hill simulation

surface potential temperature (K)

surface vapor mixing ratio (g/kg)

surface pressure (mb)

1000.000	288.00	0.00		
0.00	288.00	0.00	10.00	0.00
50.00	288.15	0.00	10.00	0.00
100.00	288.29	0.00	10.00	0.00
150.00	288.44	0.00	10.00	0.00
200.00	288.59	0.00	10.00	0.00
250.00	288.74	0.00	10.00	0.00
300.00	288.88	0.00	10.00	0.00
350.00	289.03	0.00	10.00	0.00
400.00	289.18	0.00	10.00	0.00
450.00	289.32	0.00	10.00	0.00
500.00	289.47	0.00	10.00	0.00
550.00	289.62	0.00	10.00	0.00
600.00	289.77	0.00	10.00	0.00
650.00	289.92	0.00	10.00	0.00
700.00	290.06	0.00	10.00	0.00
750.00	290.21	0.00	10.00	0.00
800.00	290.36	0.00	10.00	0.00
850.00	290.51	0.00	10.00	0.00
900.00	290.66	0.00	10.00	0.00
950.00	290.80	0.00	10.00	0.00
1000.00	290.95	0.00	10.00	0.00

Each successive line is a point in the sounding

height (m)

V (south-north) velocity (m/s)

potential temperature (K)

input_sounding

U (west-east) velocity (m/s)

Introduction

During initialization:

- Idealize test specifics are enabled in the code using the Fortran CASE construct.

```
SELECT CASE ( model_config_rec%ideal_case )
CASE ( hill2d_x )
```

- Model levels are set within the initialization to produce a stretched η coordinate (close to equally spaced z), or equally spaced η coordinate.
- Terrain is set in the initialization code
- A single sounding (z, q, Q_v, u and v) is read in from

```
WRF/test/em_hill2d_x/input_sounding
```

- Sounding is interpolated to the ARW grid, equation of state and hydrostatic balance used to compute the full thermodynamics state.
- Wind fields are interpolated to model η levels.

Introduction

• Flow over a hill • WRF Model

Note:

3D meshes are always used, even in 2D (x,z; y,z) cases. The third dimension contains only *5* planes, the boundary conditions in that dimension are periodic, and the solutions on the planes are identical in the initial state and remain so during the integration.

Introduction

• Flow over a hill • WRF Model: 2D hill simulation

After compiling, we are ready to run the case.

Set initial condition by running `ideal.exe`. Check `ideal.log` file:

```
wrf: SUCCESS COMPLETE IDEAL INIT
```

Then run wrf via run_wrf.sh

```
sbatch run_wrf.sh
```

After simulation has finished you should see a `wrfout*` file. Check `wrf.log` file

```
wrf: SUCCESS COMPLETE WRF
```

To open the output file with ncview activate conda environment:

```
conda activate env_name
ncview wrfout*
```

Here it is required to have installed miniconda, created a conda environment and installed necessary packages (**ncl**, **ncview**). If this is not the case, go to the end of this presentation to see how to do that.

To visualize all the variables available in a netcdf file

```
ncdump -h wrfout
```

Part 0 – test case em_hill2d_x

Objective:

- Become familiar with the base-case input files and results
- Run the base case and interpret the results (calculate the parameters shown on Slide 29)
- Investigate what happens when the wind speed, U, changes

Note:

Nonlinear effects usually take time to develop. Therefore, run longer simulations when using lower wind speeds (e.g. try 10x slower wind => 10x more time).

Part 0 – test case em_hill2d_x

Domain setup

$L_x = 402\ km$
$\Delta x = \Delta y = 2\ km$

41 vertical levels
$U = 10\ m/s$
$N = 0.01\ 1/s$
$h = 100\ m$
$2b = 20\ km$

- Lineary parameter

$$\hat{h} = Nh/U = 0.1$$

Linear solution

- Hydrostatic parameter

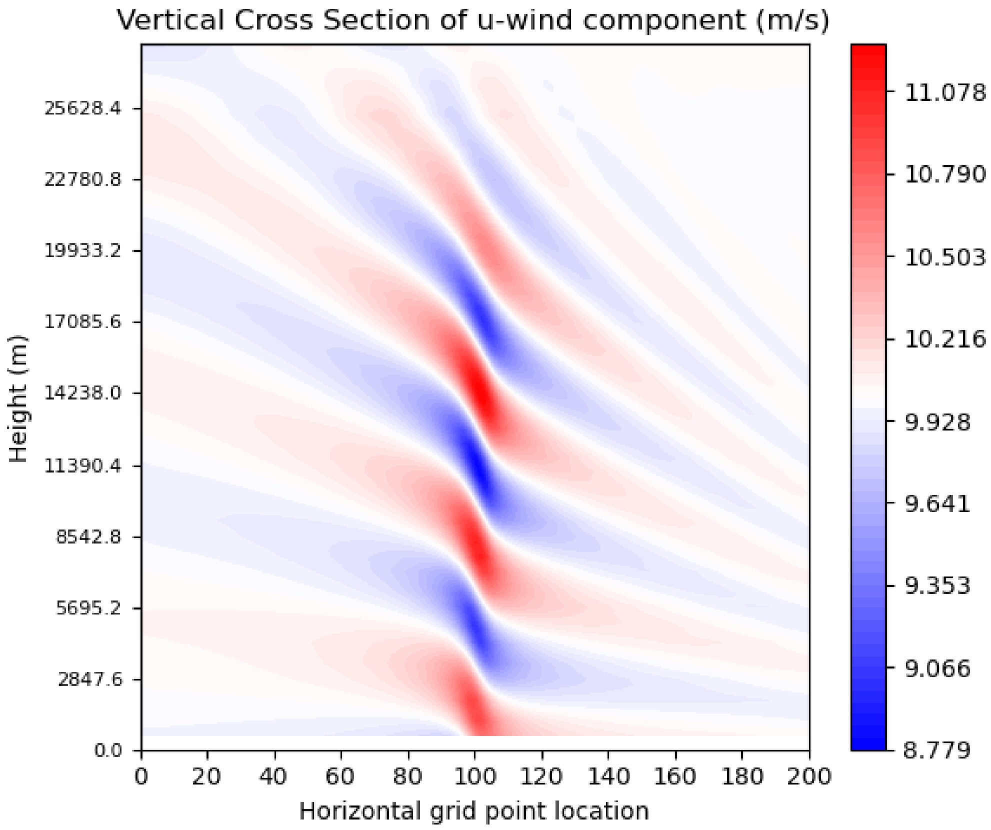
$$\hat{b} = \frac{Nb}{U} = 10 \gg 1$$

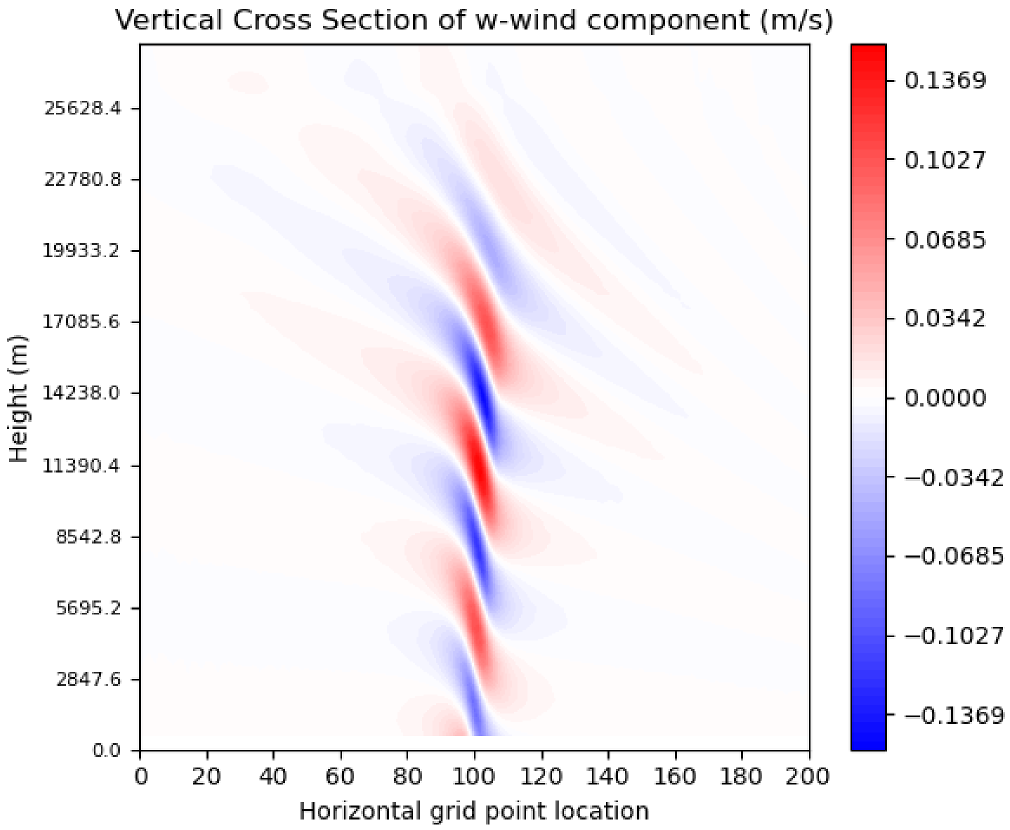
Hydrostatic mountain waves

Part 0 – test case em_hill2d_x

Hydrostatic mountain waves

$U/Nh = 10$

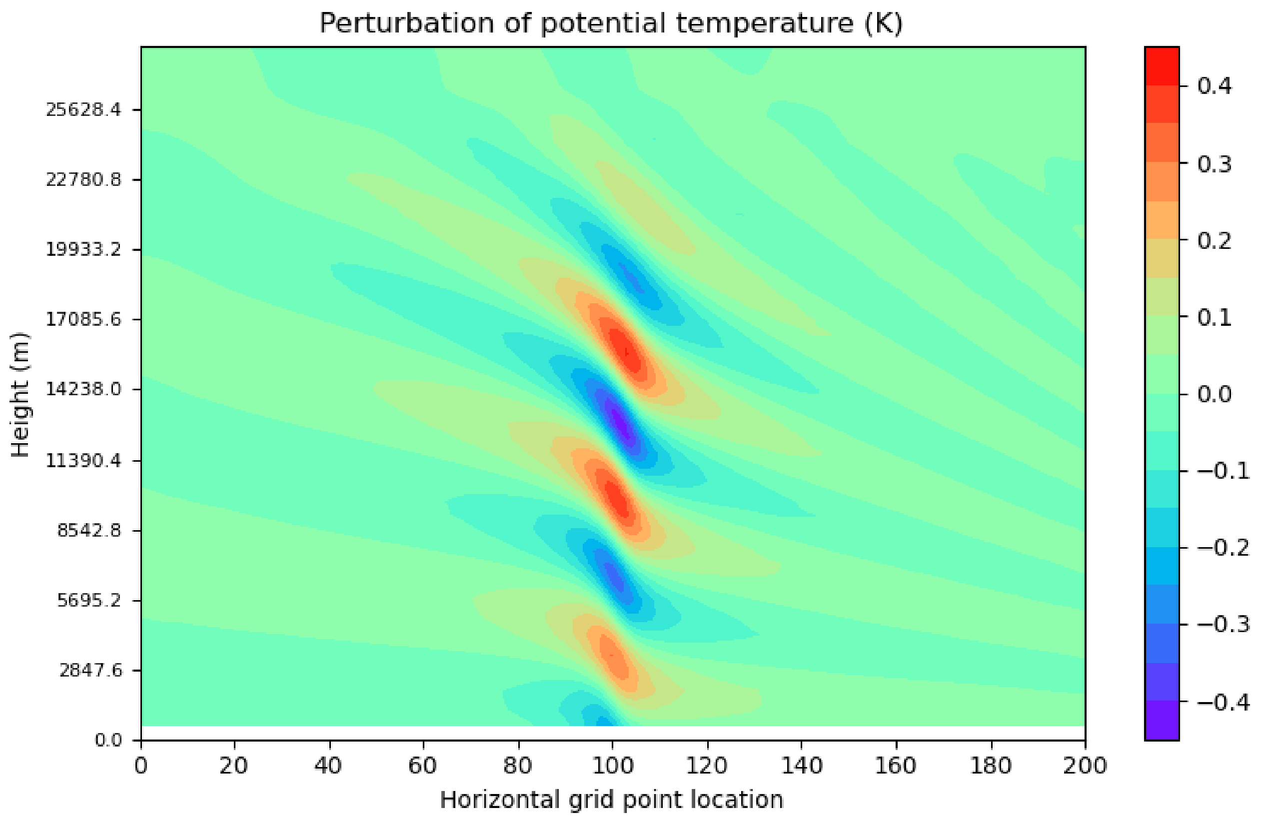




Part 0 – test case em_hill2d_x

The hill forces rising and sinking motions in a stably stratified flow, generating stationary gravity waves.

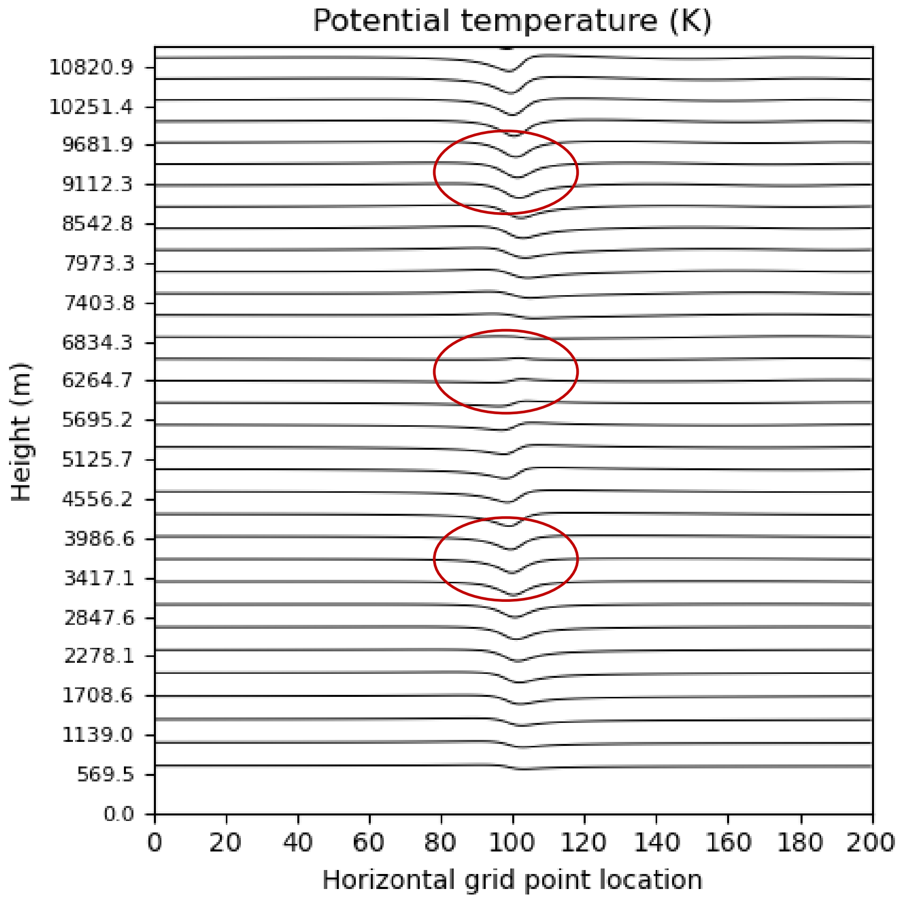
The disturbance is confined in the horizontal above the mountain, but repeats itself in vertical.



$$U/Nh = 10$$

Part 0 – test case em_hill2d_x

$\mathrm{U}/Nh = 10$



Hydrostatic mountain waves

Isentropes are displaced upward and downward in a repeating vertical pattern with a wavelength of about 6 km.

The mountain profile is reproduced at every level that is an integral multiple of the wavelength λ, while those at 3 and 9 km are roughly the mirror-image of the topography.

$$\lambda = \frac{2\pi U}{N} \cong 6km$$

Part 0 – test case em_hill2d_x

Domain setup

$L_x = 402\ km$
$\Delta x = \Delta y = 2\ km$

41 vertical levels
$U = 1\ m/s$ ← Decreasing U
$N = 0.01\ 1/s$
$h = 100\ m$
$2b = 20\ km$

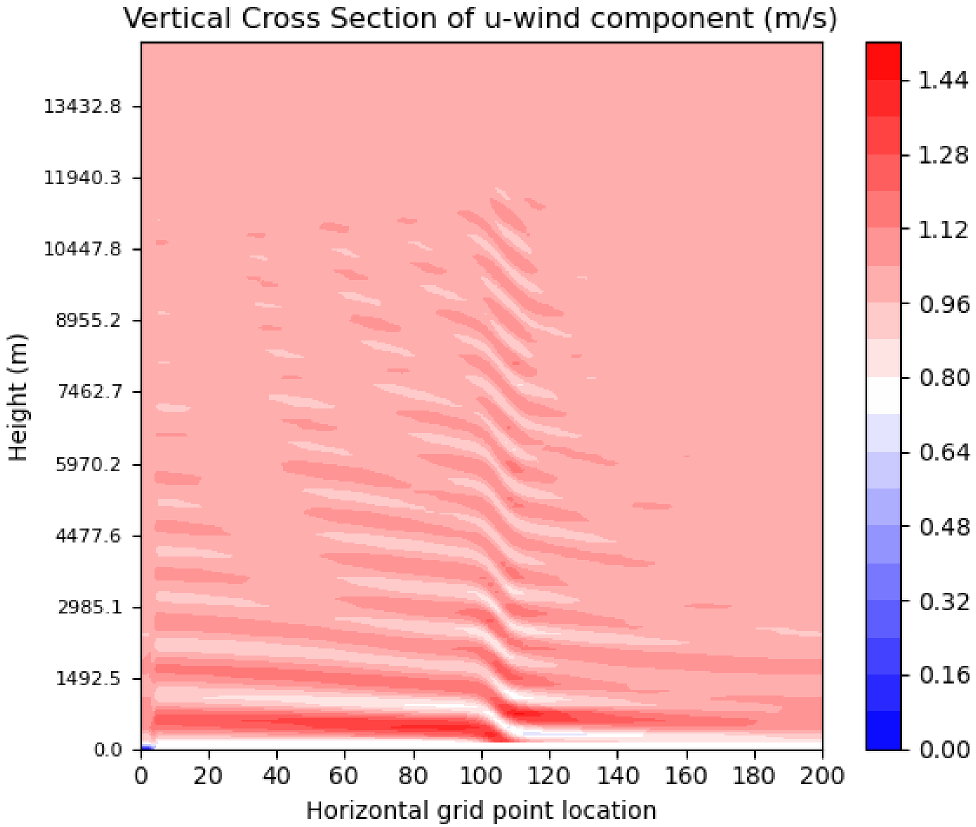
- Lineary parameter
$$\hat{h} = Nh/U = 1$$
Nonlinear effects
- Hydrostatic parameter
$$\hat{b} = \frac{Nb}{U} = 10 \gg 1$$
Hydrostatic mountain waves

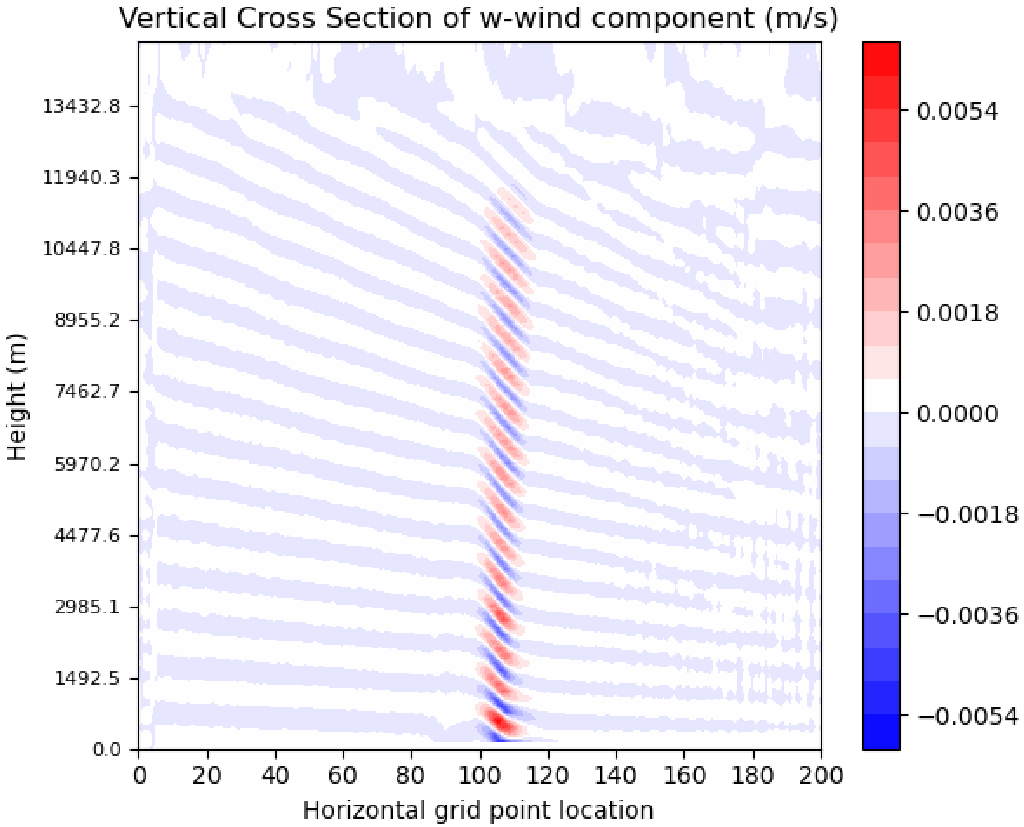
Part 0 – test case em_hill2d_x

Nonlinear effects

Slower wind -> shorter
λ -> narrower bands

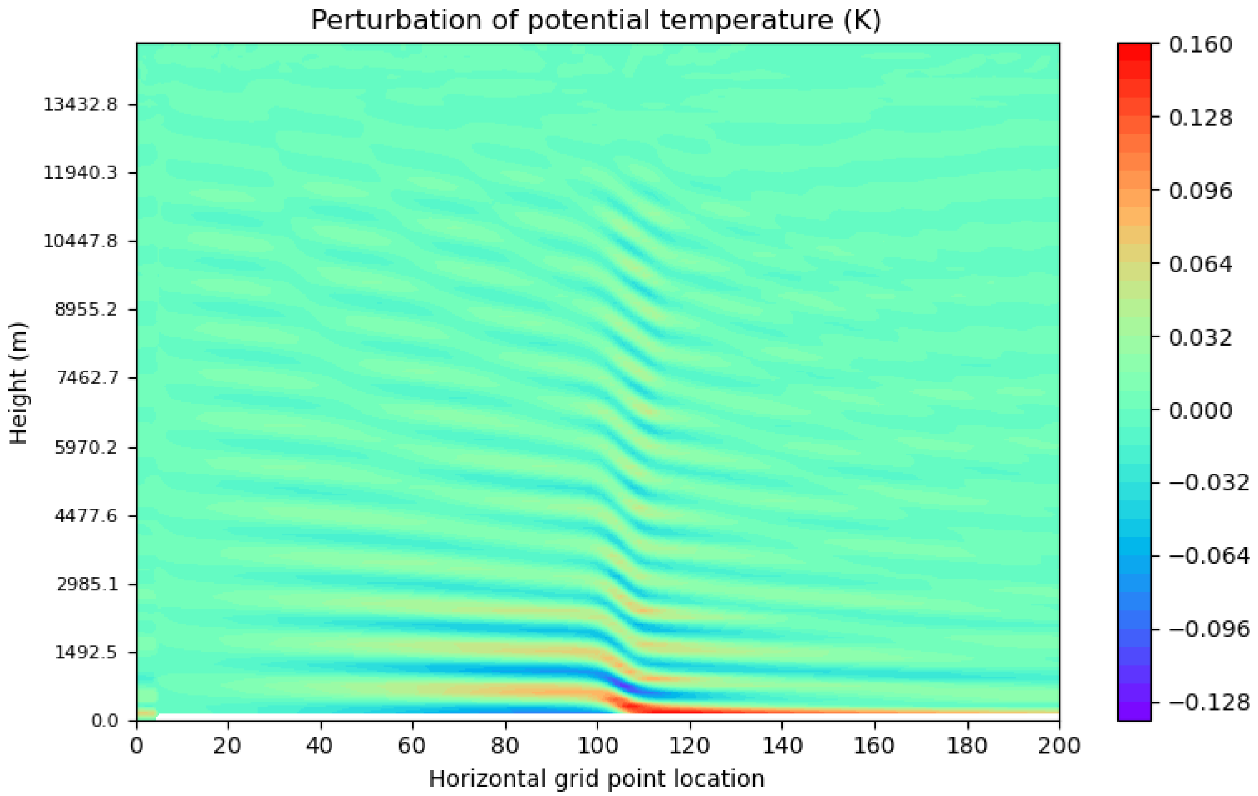
$\mathrm{U}/Nh = 1$





Part 0 – test case em_hill2d_x

When the wind is weak (U = 1 m/s), the mountain has a much stronger blocking effect. The wave pattern shortens vertically, becomes concentrated near the surface, and the flow accelerates strongly on the lee side, producing narrow bands and a flow structure that weakens with height. This marks a transition from freely propagating mountain waves to a more nonlinear, hydraulically influenced flow.



$\mathrm{U}/Nh = 1$

Soundings and Plotting

- How to change the sounding? One way is to use an [excel worksheet](#).

- For plotting we can use [wrf-python](#).

In VSCode, follow this [guide](#) to see how to create a new Jupyter notebook:

> Create a new Jupyter Notebook: ctrl+shift+p (or creating a .ipynb file)

> Select a kernel (top right)

> Create a conda environment (with a python version <= 3.11)

> Install libraries (`%conda install metpy`)

Terminal commands

- Access the cluster

```
ssh -X -J nome.cognomen#@137.204.50.15 nome.cognomen#@137.204.50.71
```

- Request an interactive job (frontend)

```
salloc -N 1 --cpus-per-task=4 --time=00:30:00 --mem=32G --constraint=blade --
partition=b1
```

```
srun --pty --overlap --jobid $JOBID bash
```

- Submit job via sbatch

```
sbatch script_name.sh
```

In here remember to modify the bash script and insert your email at line 58

- Control job queue

```
squeue –u nome.cognomen#
```

or

```
slurmtop
```

To exit from slurmtop press `ctrl+c`

Terminal commands

XXXX is the number of an open port (e.g. 2222,4321,…)

- Download/Upload file from/in the cluster

Open a window terminal and write:

```
ssh -LXXXX:137.204.50.71:22 -N nome.cognome@137.204.50.15
```

To access the cluster open another terminal window and insert:

```
ssh -X nome.cognome@127.0.0.1 -pXXXX
```

After you enable port forwarding, here you are accessing the port you opened

To **download** a file from the cluster, go into the local folder and insert:

```
scp -P XXXX nome.cognome@127.0.0.1:/path/of/the/file .
```

To **upload** a file in the cluster, go into the folder containing the file and type:

```
scp -P XXXX file_name nome.cognome@127.0.0.1:/path/of/the/final/folder
```

Terminal commands

Soft link a file

```
ln -sf nameofthefile softlinkname
```

To access the cluster open another terminal window and insert:

```
ssh -X nome.cognome@127.0.0.1 -pXXXX
```

To **download** a file from the cluster, go into the local folder and insert:

```
scp -P XXXX nome.cognome@127.0.0.1:/path/of/the/file .
```

To **upload** a file in the cluster, go into the folder containing the file and type:

```
scp -P XXXX file_name nome.cognome@127.0.0.1:/path/of/the/final/folder
```

Create a conda environment in the cluster

LINUX

Firstly, install miniconda. Go into main folder and type:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O
~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

Then, you can create a conda environment:

```
conda create -n env_name python=3.8
```

To activate the environment:

```
conda activate env_name
```

To install packages, activate conda environment first and then install:

```
conda install -c conda-forge package_name
```

Install ncl and ncview.

Create a conda environment in the cluster

For macOS:

```
brew install --cask miniconda
```

Firstly, install miniconda. Go into main folder and type:

```
mkdir -p ~/miniconda3
wget https://repo.anaconda.com/miniconda/Miniconda3-latest-Linux-x86_64.sh -O
~/miniconda3/miniconda.sh
bash ~/miniconda3/miniconda.sh -b -u -p ~/miniconda3
rm ~/miniconda3/miniconda.sh
```

Then, you can create a conda environment:

```
conda create -n env_name python=3.8
```

To activate the environment:

```
conda activate env_name
```

To install packages, activate conda environment first and then install:

```
conda install -c conda-forge package_name
```

Install ncl and ncview.

References

Lin, Y., and T. Wang, 1996: Flow Regimes and Transient Dynamics of Two-Dimensional Stratified Flow over an Isolated Mountain Ridge. J. Atmos. Sci., 53, 139–158, https://doi.org/10.1175/1520-0469(1996)053<0139:FRATDO>2.0.CO;2

Durran, D.R., 2015. Lee Waves and Mountain Waves. In: Gerald R. North (editor-in-chief), John Pyle and Fuqing Zhang (editors). Encyclopedia of Atmospheric Sciences, 2nd edition, Vol 4, pp. 95–102.

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Reinecke, P. A., and D. R. Durran, 2008: Estimating Topographic Blocking Using a Froude Number When the Static Stability Is Nonuniform. *J. Atmos. Sci.*, **65**, 1035–1048, https://doi.org/10.1175/2007JAS2100.1.

Stull, R., 2017: "Practical Meteorology: An Algebra-based Survey of Atmospheric Science" -version 1.02b. Univ. of British Columbia. 940 pages. isbn 978-0-88865-283-6.

References

WRF-related:

[WRF Technical notes (v4)]

WRF User Guide

WRF ARW User's Page

WRF GitHub Repository

Ncview features

NCL installation

Other useful stuff:

Vim cheatsheet

OPH Guide

Lab folder

Introduction

• Flow over a hill • WRF Model

If needed WRF folder can be copied from:

```
cp -r /scratch/atmos/simona.rinaldi7/complex_terrain_lab/WRFv4.6/WRF .
```